

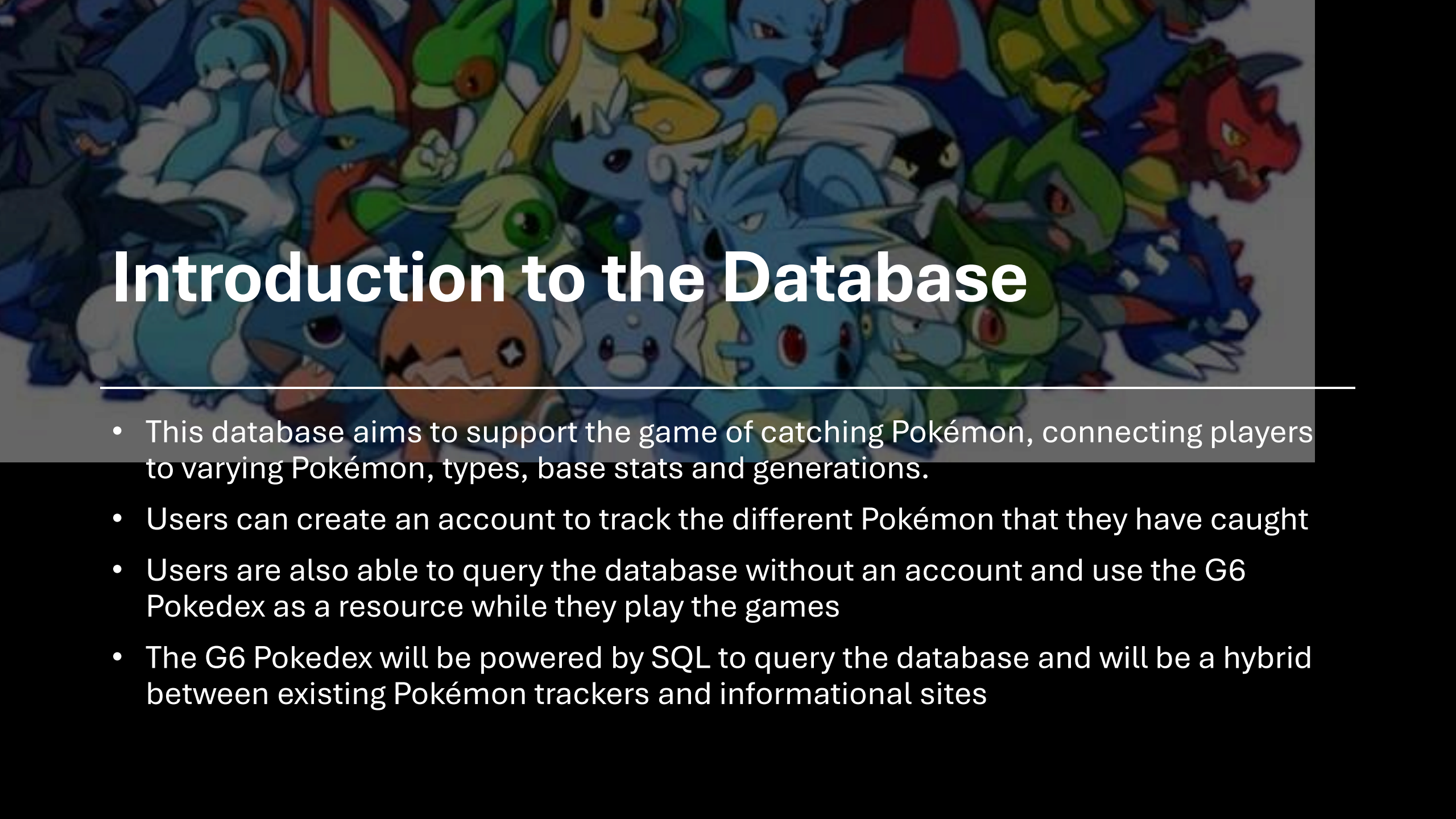


G6 Pokedex

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Introduction to the Database

- This database aims to support the game of catching Pokémon, connecting players to varying Pokémon, types, base stats and generations.
- Users can create an account to track the different Pokémon that they have caught
- Users are also able to query the database without an account and use the G6 Pokedex as a resource while they play the games
- The G6 Pokedex will be powered by SQL to query the database and will be a hybrid between existing Pokémon trackers and informational sites

Agenda

- Business requirements
- Data Analysis of the Facts
- Conceptual Model
- Logical Model
- Functionality
- Application Screen Layouts
- Concluding Thoughts



Business Requirements

- Provide a user-friendly platform to connect users to their Pokémon

- Provide a way to easily search for Pokémon by different attributes

- Provide a way for users to save Pokémon to a watchlist that can be easily accessed

- Provide a place to store Pokémon caught in games and keep track of user's progress

Data Analysis of the Facts

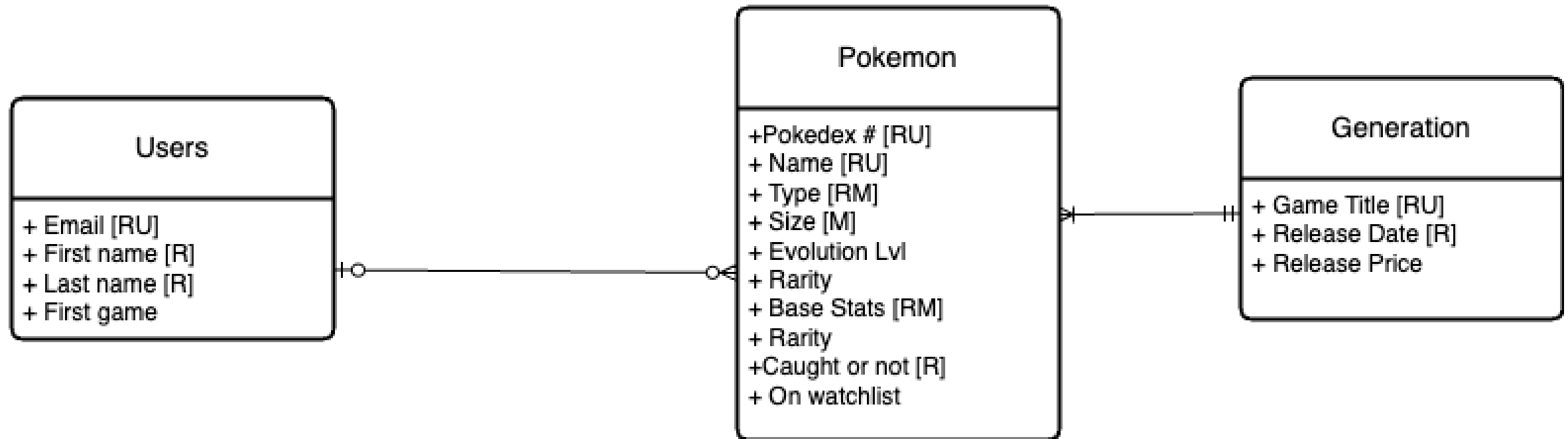
A data analysis of the facts listing the entities, attributes, and relationships in the data model.

Entity	Attribute	Props	Description
<u>Pokemon</u>	Pokedex #	RU	Unique identification # for each Pokemon
	Name	RU	Name of Pokemon
	Type	RM	All types of a Pokemon
	Size	RM	Height and weight of Pokemon
	Evolution Level	R	Level Pokemon evolves at
	Capture Rate	R	Base rate to catch Pokemon with no damage done
	Base Stats	RM	HP, speed, physical/special attack, physical/special defense of Pokemon
	Rarity		How rare a Pokemon is to find
	Caught	R	Has the user caught it?
	Watchlist		Add Pokemon to watch list
<u>User</u>	First Name	R	
	Last Name	R	
	Email	RU	
<u>Generations</u>	Game Title	RU	Game title generation was introduced in
	Game Release Date	R	Date of generation release
	Release Price		Price of game at release

Relationship	Entity	Rules	M/n	Max	Entity 2
Pokemon introduced in generation	Pokemon	Is introduced in		1 1	Generation
	Generation	Introduces		1 M	Pokemon
User and Pokemon	User	Has		0 M	Pokemon
	Pokemon	Has		0 1	User

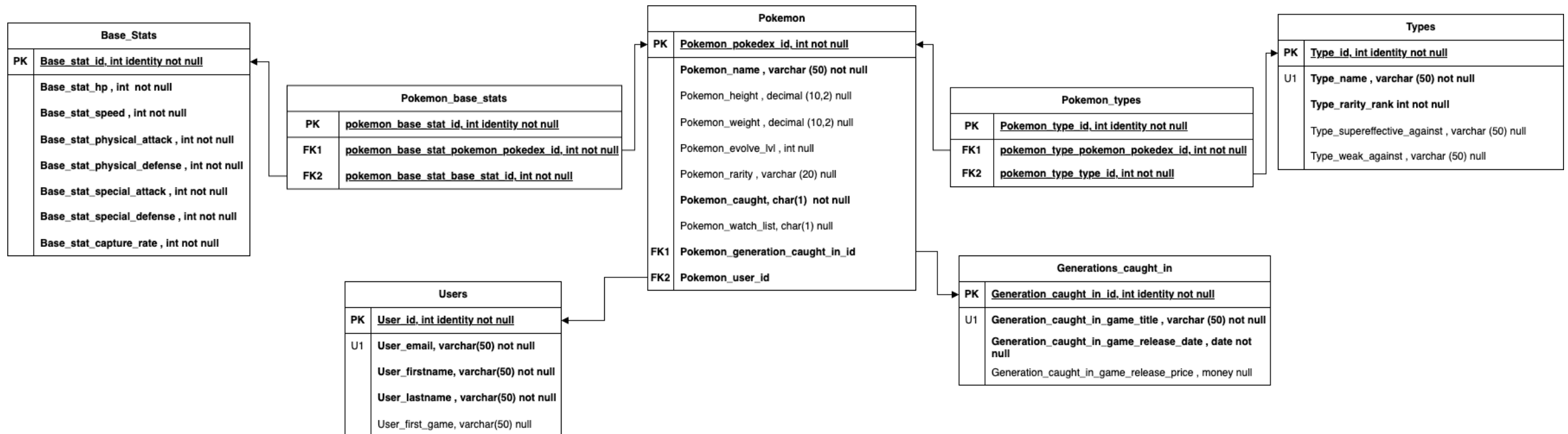
Conceptual Model

The Pokémon entity has relationships to both the User entity and the Generation entity.

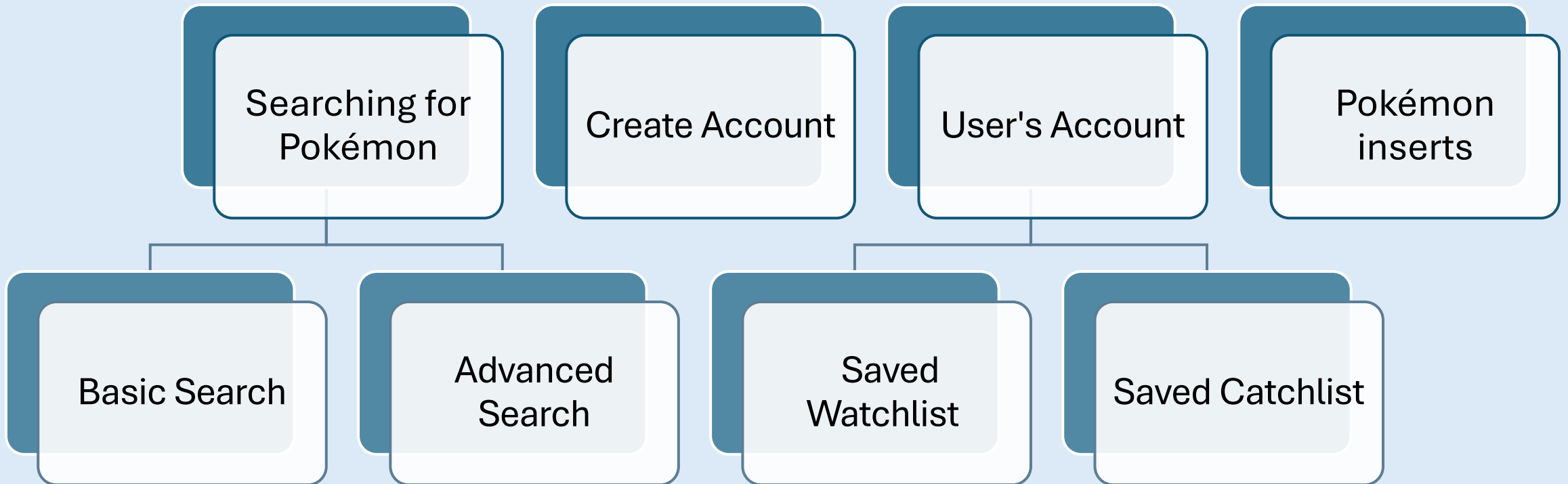


Logical Model

Entities, attributes, and relationships are identified to display the needed data and business logic. Bridge tables were created, and Primary Keys/Foreign keys were used to ensure the integrity of the data.



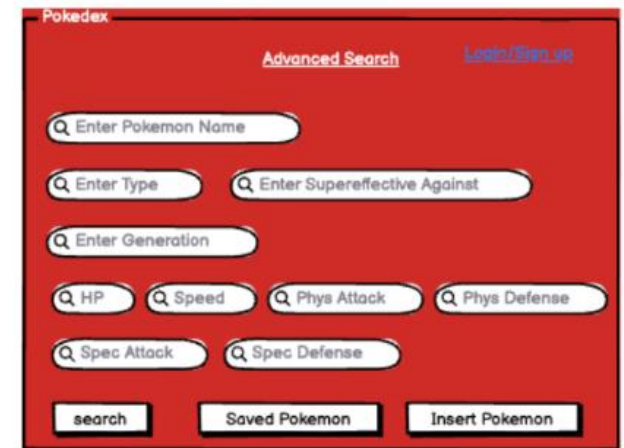
Functionality Visualization



Functionality

- Questions from report (no screenshots)

Application Screen Layouts



Home page and advanced search screens

Application Screen Layouts



User Login

< Back

Username

Password

Sign in

This is a wireframe for a 'User Login' screen. It features a red background. At the top left, there is a title 'User Login' and a '< Back' button. In the center, there are two input fields labeled 'Username' and 'Password'. At the bottom center, there is a 'Sign in' button.



Create Account

< Back

Username

Email Address

Password

First Name

Re-enter Password

Last Name

First Pokemon Game?

Create Account

This is a wireframe for a 'Create Account' screen. It features a red background. At the top left, there is a title 'Create Account' and a '< Back' button. Below the back button, there are six input fields arranged in two columns: 'Username', 'Email Address', 'Password', 'First Name', 'Re-enter Password', and 'Last Name'. At the bottom, there is a 'First Pokemon Game?' input field and a 'Create Account' button.

User account sign in and create account screens

Application Screen Layouts

Pokemon

Home

GenerationTypesBase Stats

Pokemon

Saved Pokemon	Pokedex ID	Pokemon Name	Height (ft/in)	Weight (Lbs)	Evolve Lvl	Caught?	Generation Game Title
Save	001	Bulbasaur	2.04	15.2	16	☐ Checkbox	Red/Blue

Type

Home

PokemonGenerationBase Stats

Type

Type Name	Supereffective against	Weak against
Grass	Electric/Water/Ground	Fire/Ice/Poison/Flying/Bug
Poison	Grass/Fight/Poison/Bug	Ground/Psychic

Generation

Home

PokemonTypesBase Stats

Generation

Generation Caught In	Game Title	Game Release Date (North America)	Game Release Price
01	Red/Blue	04-28-1998	\$24.95

Base Stats

Home

PokemonGenerationTypes

Base Stats

Pokemon Name	HP	Speed	Physical Attack	Physical Defense	Special Attack	Special Defense	Base Catch Rate
Bulbasaur	45	45	49	49	65	65	45

Saved Pokemon

Home

GenerationTypesBase Stats

Saved Pokemon

Saved Pokemon	Pokedex ID	Pokemon Name	Height (ft/in)	Weight (Lbs)	Evolve Lvl	Caught?	Generation Game Title
Save	002	Ivysaur	3.03	28.7	32	☐ Checkbox	Red/Blue
Save	003	Venasaur	6.07	220.5	NA	☒ Checkbox	Red/Blue

Enter Pokemon

Home

Insert Pokemon
--If value unknown, leave blank--

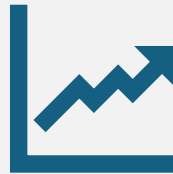
Pokedex ID	Pokemon Name	Height (ft/in)	Weight (Lbs)	Evolve Lvl	Type Name	HP	Speed	Physical Attack	Physical Defense	Special Attack	Special Defense
004	Charmander			16	Fire						

Output of all tables, saved Pokémon watchlist, and insert Pokémon screens

Limitations



INDEXING NEEDS TO BE ADDED TO INCREASE
THE EFFICIENCY OF THE DATABASE



POPULATING THE
DATABASE CAN IMPACT STORAGE
CAPABILITIES AND OVERALL PERFORMANCE



SCALABILITY MAY REQUIRE SERVERS FOR
RUNNING LARGER DATABASES WHICH CAN
BE PROBLEMATIC

Conclusion

With so many different Pokémon in existence having so many different attributes, it is currently very difficult to search for Pokémon by these attributes. It is very surprising that the most popular database currently for Pokémon, serebii.net, doesn't have this feature available to users already. Following the implementation of our database, Pokémon research becomes much more streamlined. Team 6 created the G6 Pokedex database with the user in mind, and the G6 Pokedex gives users the edge they need to win more battles, catch more Pokémon, and be the very best that no one ever was.

